Advanced Microeconomics: Solutions to Problems 3 and 5

CES Production, Profit Maximisation and Cost Minimisation; Insurance under Asymmetric Information

Problem 3: CES Production Function

Given. The three-input CES production function is

$$Y = A\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)^{v/\rho}, A>0, \alpha_i>0, \sum_i \alpha_i = 1, 0 \neq \rho \leq 1, v>0.$$

The output price is p and input prices are w_1, w_2, w_3. To keep the algebra compact, define the inner aggregate and the substitution elasticity:

$$S \equiv \alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho} \Rightarrow Y = A\, S^{v/\rho}, \sigma \equiv \frac{1}{1-\rho}.$$

Some exponent identities used repeatedly below:

$$\sigma\,\rho = \frac{\rho}{1-\rho} = \sigma - 1, \frac{\sigma - 1}{\rho} = \sigma, 1 - \sigma = \frac{-\rho}{1-\rho}, \frac{1}{\rho - 1} = -\sigma.$$

(a) Returns to Scale

Step 1: Multiply every input by the same factor $t>0$.

$$Y\left(t X_1, t X_2, t X_3\right) = A\left(\alpha_1 \left(t X_1\right)^{\rho} + \alpha_2 \left(t X_2\right)^{\rho} + \alpha_3 \left(t X_3\right)^{\rho}\right)^{v/\rho}.$$

Step 2: Use $\left(t X_i\right)^{\rho} = t^{\rho} X_i^{\rho}$ and factor out t^{ρ}.

$$Y\left(t X_1, t X_2, t X_3\right) = A\left(t^{\rho}\alpha_1 X_1^{\rho} + t^{\rho}\alpha_2 X_2^{\rho} + t^{\rho}\alpha_3 X_3^{\rho}\right)^{v/\rho} = A\left[t^{\rho}\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)\right]^{v/\rho}.$$

Step 3: Apply the power to each factor, $\left(a\,b\right)^k = a^k b^k$.

$$Y\left(t X_1, t X_2, t X_3\right) = \left(t^{\rho}\right)^{v/\rho} \cdot A\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)^{v/\rho} = t^{\rho \cdot v/\rho} Y = t^{v} Y\left(X_1, X_2, X_3\right).$$

Step 4: Compare with the definition of returns to scale. A function is homogeneous of degree k if $f\left(t X\right) = t^k f\left(X\right)$. Here $k = v$. For $t>1$ (expanding all inputs):

- if $v>1$, then $t^{v}>t$, so output rises more than proportionally;
- if $v=1$, then $t^{v}=t$, so output rises exactly proportionally;
- if $v<1$, then $t^{v}<t$, so output rises less than proportionally.

Illustration: doubling all inputs ($t=2$) with $v=1.2$ gives $2^{1.2} \approx 2.30$ times the output. With $v=0.8$ it gives $2^{0.8} \approx 1.74$ times the output.

Answer (a): The CES function is homogeneous of degree ν. If $\nu > 1$ there are increasing returns to scale, if $\nu = 1$ constant returns to scale, and if $\nu < 1$ decreasing returns to scale. The returns to scale do not depend on A, α_i or ρ.

(b) Marginal Rate of Technical Substitution

Step 1: Set up the chain rule. Write $Y = A\,S^{\nu/\rho}$, where S depends on X_i. Then

$$\frac{\partial Y}{\partial X_i} = \frac{dY}{dS}\cdot\frac{\partial S}{\partial X_i}, \text{ where } \frac{dY}{dS} = A\,\frac{\nu}{\rho}\,S^{\frac{\nu}{\rho}-1}.$$

Step 2: Inner derivative. Only the i-th term of S contains X_i:

$$\frac{\partial S}{\partial X_i} = \frac{\partial}{\partial X_i}\left(\alpha_i X_i^{\rho}\right) = \rho\,\alpha_i X_i^{\rho-1}.$$

Step 3: Combine and simplify. The ρ in the numerator cancels the ρ in the denominator, and $\frac{\nu}{\rho} - 1 = \frac{\nu-\rho}{\rho}$:

$$\mathrm{MP}_i = A\,\frac{\nu}{\rho}\,S^{\frac{\nu-\rho}{\rho}}\cdot\rho\,\alpha_i X_i^{\rho-1} = \nu\,A\,\alpha_i X_i^{\rho-1} S^{\frac{\nu-\rho}{\rho}}.$$

Written out for each input:

$$\mathrm{MP}_1 = \nu\,A\,\alpha_1 X_1^{\rho-1} S^{\frac{\nu-\rho}{\rho}},\ \mathrm{MP}_2 = \nu\,A\,\alpha_2 X_2^{\rho-1} S^{\frac{\nu-\rho}{\rho}},\ \mathrm{MP}_3 = \nu\,A\,\alpha_3 X_3^{\rho-1} S^{\frac{\nu-\rho}{\rho}}.$$

Step 4: Equivalent form in terms of Y. From $Y = A\,S^{\nu/\rho}$ we get $S = (Y/A)^{\rho/\nu}$, so

$$S^{\frac{\nu-\rho}{\rho}} = \left(\frac{Y}{A}\right)^{\frac{\rho}{\nu}\cdot\frac{\nu-\rho}{\rho}} = \left(\frac{Y}{A}\right)^{\frac{\nu-\rho}{\nu}} \Rightarrow \mathrm{MP}_i = \nu\,\alpha_i A^{\rho/\nu} Y^{1-\rho/\nu} X_i^{\rho-1} > 0.$$

Step 5: Take the ratio of marginal products. The common factor $\nu\,A\,S^{(\nu-\rho)/\rho}$ appears in both and cancels:

$$\mathrm{MRTS}_{ij} = \frac{\mathrm{MP}_i}{\mathrm{MP}_j} = \frac{\nu\,A\,\alpha_i X_i^{\rho-1} S^{\frac{\nu-\rho}{\rho}}}{\nu\,A\,\alpha_j X_j^{\rho-1} S^{\frac{\nu-\rho}{\rho}}} = \frac{\alpha_i}{\alpha_j}\cdot\frac{X_i^{\rho-1}}{X_j^{\rho-1}} = \frac{\alpha_i}{\alpha_j}\left(\frac{X_i}{X_j}\right)^{\rho-1} = \frac{\alpha_i}{\alpha_j}\left(\frac{X_j}{X_i}\right)^{1-\rho}.$$

Step 6: The three pairwise MRTS values.

$$\mathrm{MRTS}_{12} = \frac{\alpha_1}{\alpha_2}\left(\frac{X_2}{X_1}\right)^{1-\rho},\ \mathrm{MRTS}_{13} = \frac{\alpha_1}{\alpha_3}\left(\frac{X_3}{X_1}\right)^{1-\rho},\ \mathrm{MRTS}_{23} = \frac{\alpha_2}{\alpha_3}\left(\frac{X_3}{X_2}\right)^{1-\rho}.$$

Step 7: Slope behaviour along an isoquant. Differentiating $\ln \mathrm{MRTS}_{ij}$ with respect to $\ln(X_i/X_j)$ gives $-(1-\rho)$. This is negative when $\rho<1$, so the MRTS falls as X_i replaces X_j (diminishing MRTS, convex isoquants). When $\rho=1$ it is zero, so the MRTS is constant.

Answer (b): $\mathrm{MRTS}_{ij}=\dfrac{\alpha_i}{\alpha_j}\left(\dfrac{X_j}{X_i}\right)^{1-\rho}$. It depends only on the ratio of the two inputs, not on A, ν or the third input, so the technology is homothetic and each pair of inputs is weakly separable from the third. For $\rho<1$ the MRTS falls as X_i/X_j rises, so isoquants are convex. For $\rho=1$ the MRTS equals the constant α_i/α_j, so isoquants are straight lines.

(c) Elasticity of Substitution

Step 1: Definition. The elasticity of substitution between inputs i and j is the percentage change in the input ratio for a one per cent change in the MRTS:

$$\sigma_{ij}=\frac{d\ln(X_j/X_i)}{d\ln \mathrm{MRTS}_{ij}}.$$

Step 2: Take logs of the MRTS from part (b).

$$\ln \mathrm{MRTS}_{ij}=\ln\alpha_i-\ln\alpha_j+(1-\rho)\ln\frac{X_j}{X_i}.$$

Step 3: Differentiate with respect to $\ln(X_j/X_i)$. The terms $\ln\alpha_i$ and $\ln\alpha_j$ are constants:

$$\frac{d\ln \mathrm{MRTS}_{ij}}{d\ln(X_j/X_i)}=0-0+(1-\rho)=1-\rho.$$

Step 4: Invert to get σ_{ij}.

$$\sigma_{ij}=\frac{1}{d\ln \mathrm{MRTS}_{ij}/d\ln(X_j/X_i)}=\frac{1}{1-\rho}.$$

Because the same calculation applies to every pair, $\sigma_{12}=\sigma_{13}=\sigma_{23}=1/(1-\rho)$.

Step 5: Equivalent derivation from the inverted relationship. Solving $\mathrm{MRTS}_{ij}=(\alpha_i/\alpha_j)(X_j/X_i)^{1-\rho}$ for the input ratio:

$$\frac{X_j}{X_i}=\left(\frac{\alpha_j}{\alpha_i}\mathrm{MRTS}_{ij}\right)^{\frac{1}{1-\rho}} \Rightarrow \ln\frac{X_j}{X_i}=\frac{1}{1-\rho}\ln\frac{\alpha_j}{\alpha_i}+\frac{1}{1-\rho}\ln \mathrm{MRTS}_{ij}.$$

The coefficient on $\ln \mathrm{MRTS}_{ij}$ is $1/(1-\rho)$, confirming Step 4.

Step 6: Special cases.

Value of ρ	$\sigma = 1/(1-\rho)$	Technology
$\rho \to 1$	$\sigma \to \infty$	Perfect substitutes (linear)
$0<\rho<1$	$\sigma>1$	Easier to substitute than Cobb–Douglas
$\rho \to 0$	$\sigma \to 1$	Cobb–Douglas
$\rho<0$	$0<\sigma<1$	Harder to substitute than Cobb–Douglas
$\rho \to -\infty$	$\sigma \to 0$	Perfect complements (Leontief)

Illustration: $\rho = 0.5$ gives $\sigma = 2$, and $\rho = -1$ gives $\sigma = 0.5$.

Answer (c): $\sigma_{12} = \sigma_{13} = \sigma_{23} = \sigma = \dfrac{1}{1-\rho}$. The elasticity of substitution is constant (hence "constant elasticity of substitution"), identical for every pair of inputs, and independent of input levels, A, α_i and v.

(d) Profit Function and First-Order Conditions

Step 1: Revenue and cost.

$$\text{TR} = pY = pA\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)^{v/\rho}, \quad \text{TC} = w_1 X_1 + w_2 X_2 + w_3 X_3.$$

Step 2: The profit maximisation problem.

$$\max_{X_1, X_2, X_3} \pi = pA\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)^{v/\rho} - w_1 X_1 - w_2 X_2 - w_3 X_3.$$

Step 3: First-order conditions. Differentiate with respect to each input and use MP_i from part (b):

$$\frac{\partial \pi}{\partial X_1} = pvA\alpha_1 X_1^{\rho-1} S^{\frac{v-\rho}{\rho}} - w_1 = 0, \quad (\text{FOC}_1)$$

$$\frac{\partial \pi}{\partial X_2} = pvA\alpha_2 X_2^{\rho-1} S^{\frac{v-\rho}{\rho}} - w_2 = 0, \quad (\text{FOC}_2)$$

$$\frac{\partial \pi}{\partial X_3} = pvA\alpha_3 X_3^{\rho-1} S^{\frac{v-\rho}{\rho}} - w_3 = 0. \quad (\text{FOC}_3)$$

In compact form, $p \cdot \text{MP}_i = w_i$: each input is hired until the value of its marginal product equals its price.

Step 4: Implication of the FOCs (input ratios). Dividing FOC_1 by FOC_2 and FOC_1 by FOC_3, the terms $pvAS^{(v-\rho)/\rho}$ cancel:

$$\frac{w_1}{w_2}=\frac{\alpha_1}{\alpha_2}\left(\frac{X_1}{X_2}\right)^{\rho-1}=\mathrm{MRTS}_{12},\quad \frac{w_1}{w_3}=\frac{\alpha_1}{\alpha_3}\left(\frac{X_1}{X_3}\right)^{\rho-1}=\mathrm{MRTS}_{13}.$$

So at the optimum the MRTS between any two inputs equals their price ratio.

Step 5: Second-order condition. An interior maximum requires π to be strictly concave in (X_1, X_2, X_3). A CES function with $\rho<1$ is quasi-concave and homogeneous of degree v, so it is strictly concave when $v<1$. If $v\geq 1$, scaling all inputs by $t>1$ multiplies revenue by $t^v \geq t$ but cost only by t, so any positive profit can be increased without limit. Profit is then either unbounded or zero, and there is no finite interior maximum. **From here on we assume $v<1$ (decreasing returns) and $\rho<1$.**

Answer (d): $\pi = p\,A\left(\sum_i \alpha_i X_i^{\rho}\right)^{v/\rho} - \sum_i w_i X_i$. The FOCs are $p\,v\,A\,\alpha_i X_i^{\rho-1} S^{(v-\rho)/\rho} = w_i$ for $i=1,2,3$, meaning the value of each input's marginal product equals its price. The SOC holds when $v<1$.

(e) Input Demand and Output Supply Functions

Step 1: Isolate $X_i^{\rho-1}$ in each FOC.

$$X_i^{\rho-1}=\frac{w_i}{p\,v\,A\,\alpha_i S^{\frac{v-\rho}{\rho}}}.$$

Step 2: Raise both sides to the power $\frac{1}{\rho-1}=-\sigma$. A negative power flips the fraction:

$$X_i=\left(\frac{w_i}{p\,v\,A\,\alpha_i S^{\frac{v-\rho}{\rho}}}\right)^{-\sigma}=\left(\frac{p\,v\,A\,\alpha_i S^{\frac{v-\rho}{\rho}}}{w_i}\right)^{\sigma}=\left(p\,v\,A\,S^{\frac{v-\rho}{\rho}}\right)^{\sigma}\left(\frac{\alpha_i}{w_i}\right)^{\sigma}. \quad (1)$$

X_i still depends on S, which itself depends on all three X's. The next steps solve for S.

Step 3: Compute $\alpha_i X_i^{\rho}$ from (1). Raise (1) to the power ρ and multiply by α_i, using $\sigma\,\rho=\sigma-1$:

$$\alpha_i X_i^{\rho}=\alpha_i\left(p\,v\,A\,S^{\frac{v-\rho}{\rho}}\right)^{\sigma-1}\alpha_i^{\sigma-1}w_i^{-(\sigma-1)}=(p\,v\,A)^{\sigma-1}S^{\frac{(v-\rho)(\sigma-1)}{\rho}}\alpha_i^{\sigma}w_i^{1-\sigma}.$$

Since $(\sigma-1)/\rho=\sigma$, the power of S is $(v-\rho)\sigma$:

$$\alpha_i X_i^{\rho}=(p\,v\,A)^{\sigma-1}S^{(v-\rho)\sigma}\alpha_i^{\sigma}w_i^{1-\sigma}.$$

Step 4: Sum over $i=1,2,3$ to rebuild S.

$$S=\sum_i \alpha_i X_i^{\rho}=(p\,v\,A)^{\sigma-1}S^{(v-\rho)\sigma}\left(\alpha_1^{\sigma}w_1^{1-\sigma}+\alpha_2^{\sigma}w_2^{1-\sigma}+\alpha_3^{\sigma}w_3^{1-\sigma}\right).$$

Step 5: Define the input-price index W.

$$W \equiv \left(\alpha_1^{\sigma} w_1^{1-\sigma} + \alpha_2^{\sigma} w_2^{1-\sigma} + \alpha_3^{\sigma} w_3^{1-\sigma}\right)^{\frac{1}{1-\sigma}} \Leftrightarrow \sum_i \alpha_i^{\sigma} w_i^{1-\sigma} = W^{1-\sigma}.$$

In terms of ρ: $W = \left(\sum_i \alpha_i^{\frac{1}{1-\rho}} w_i^{\frac{-\rho}{1-\rho}}\right)^{-\frac{1-\rho}{\rho}}$. Two useful facts:

- W is homogeneous of degree 1 in (w_1, w_2, w_3): replacing each w_i by $t\,w_i$ multiplies the sum by $t^{1-\sigma}$, and the outer power $\frac{1}{1-\sigma}$ turns this into t.
- $\frac{\partial W}{\partial w_i} = \frac{1}{1-\sigma} W^{\sigma} \cdot (1-\sigma)\alpha_i^{\sigma} w_i^{-\sigma} = W^{\sigma} \alpha_i^{\sigma} w_i^{-\sigma} = \left(\frac{\alpha_i W}{w_i}\right)^{\sigma}$.

Step 6: Solve for S. Substituting W into Step 4:

$$S = (p\,v\,A)^{\sigma-1} S^{(v-\rho)\sigma} W^{1-\sigma} \Rightarrow S^{1-(v-\rho)\sigma} = (p\,v\,A)^{\sigma-1} W^{1-\sigma} = \left(\frac{p\,v\,A}{W}\right)^{\sigma-1}.$$

Simplify the exponent on the left using $\rho\sigma = \sigma - 1$:

$$1 - (v-\rho)\sigma = 1 - v\sigma + \rho\sigma = 1 - v\sigma + \sigma - 1 = \sigma(1-v).$$

Therefore

$$S^{\sigma(1-v)} = \left(\frac{v\,p\,A}{W}\right)^{\sigma-1} \Rightarrow S = \left(\frac{v\,p\,A}{W}\right)^{\frac{\sigma-1}{\sigma(1-v)}} = \left(\frac{v\,p\,A}{W}\right)^{\frac{\rho}{1-v}},$$

using $(\sigma-1)/\sigma = \rho$. Taking the $1/\rho$ power:

$$S^{1/\rho} = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}}.$$

Step 7: Output supply. Since $Y = A\,S^{v/\rho} = A\left(S^{1/\rho}\right)^{v}$:

$$Y^*(p,w) = A\left(\frac{v\,p\,A}{W}\right)^{\frac{v}{1-v}} = A^{1+\frac{v}{1-v}}\left(\frac{v\,p}{W}\right)^{\frac{v}{1-v}} = A^{\frac{1}{1-v}}\left(\frac{v\,p}{W}\right)^{\frac{v}{1-v}}.$$

Step 8: Input demands. Return to (1). First evaluate the bracket $p\,v\,A\,S^{(v-\rho)/\rho}$ using the solution for S:

$$S^{\frac{v-\rho}{\rho}} = \left(\frac{v\,p\,A}{W}\right)^{\frac{\rho}{1-v}\cdot\frac{v-\rho}{\rho}} = \left(\frac{v\,p\,A}{W}\right)^{\frac{v-\rho}{1-v}}.$$

$$p\,v\,A\,S^{\frac{v-\rho}{\rho}} = (v\,p\,A)^{1+\frac{v-\rho}{1-v}} W^{-\frac{v-\rho}{1-v}} = (v\,p\,A)^{\frac{1-\rho}{1-v}} W^{-\frac{v-\rho}{1-v}}.$$

Raise this to the power $\sigma = \frac{1}{1-\rho}$:

$$\left(p\,v\,A\,S^{\frac{v-\rho}{\rho}}\right)^{\sigma} = (v\,p\,A)^{\frac{1}{1-v}} W^{-\frac{v-\rho}{(1-v)(1-\rho)}}.$$

Now note $-\frac{v-\rho}{(1-v)(1-\rho)} = \frac{1}{1-\rho} - \frac{1}{1-v} = \sigma - \frac{1}{1-v}$, so $W^{-\frac{v-\rho}{(1-v)(1-\rho)}} = W^{-\frac{1}{1-v}} W^{\sigma}$. Substituting into (1):

$$X_i^*(p,w) = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} W^{\sigma} \left(\frac{\alpha_i}{w_i}\right)^{\sigma} = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} \left(\frac{\alpha_i W}{w_i}\right)^{\frac{1}{1-\rho}}.$$

For each input separately:

$$X_1^* = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} \left(\frac{\alpha_1 W}{w_1}\right)^{\frac{1}{1-\rho}}, X_2^* = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} \left(\frac{\alpha_2 W}{w_2}\right)^{\frac{1}{1-\rho}}, X_3^* = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} \left(\frac{\alpha_3 W}{w_3}\right)^{\frac{1}{1-\rho}}.$$

Step 9: Total cost at the optimum. Multiply each X_i^* by w_i and add:

$$\sum_i w_i X_i^* = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} W^{\sigma} \sum_i \alpha_i^{\sigma} w_i^{1-\sigma} = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} W^{\sigma} W^{1-\sigma} = W\,S^{1/\rho}.$$

Also $p\,Y^* = p\,A\left(S^{1/\rho}\right)^{v}$ and $\left(S^{1/\rho}\right)^{1-v} = v\,p\,A/W$, so $W\,S^{1/\rho} = v\,p\,A\left(S^{1/\rho}\right)^{v} = v\,p\,Y^*$. Hence

$$\sum_i w_i X_i^* = v\,p\,Y^*:$$

the firm spends the share v of its revenue on inputs.

Step 10: Properties of the input demand functions. Let $s_i = w_i X_i^* / \sum_j w_j X_j^* = \alpha_i^{\sigma} w_i^{1-\sigma} / W^{1-\sigma}$ denote input i's cost share. From Step 5, $\partial \ln W / \partial \ln w_i = \frac{w_i}{W}\left(\frac{\alpha_i W}{w_i}\right)^{\sigma} = s_i$.

Property 1: homogeneous of degree zero in (p,w). Replace (p,w) by $(t\,p, t\,w)$. Then $W \to t\,W$, so

$$X_i^*(t\,p, t\,w) = \left(\frac{v\,t\,p\,A}{t\,W}\right)^{\frac{1}{1-v}} \left(\frac{\alpha_i t\,W}{t\,w_i}\right)^{\sigma} = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}} \left(\frac{\alpha_i W}{w_i}\right)^{\sigma} = X_i^*(p,w). \checkmark$$

Property 2: downward-sloping in own price. Take logs of X_i^*:

$$\ln X_i^* = \frac{1}{1-v}(\ln v + \ln p + \ln A - \ln W) + \sigma(\ln \alpha_i + \ln W - \ln w_i).$$

Differentiate with respect to $\ln w_i$, using $\partial \ln W / \partial \ln w_i = s_i$:

$$\varepsilon_{ii}=\frac{\partial \ln X_i^*}{\partial \ln w_i}=-\frac{s_i}{1-v}+\sigma s_i-\sigma=-\left[\sigma\left(1-s_i\right)+\frac{s_i}{1-v}\right]<0.\checkmark$$

Both terms in the bracket are positive since $0<s_i<1$, $\sigma>0$ and $v<1$.

Property 3: cross-price effects and symmetry. Differentiate the same log expression with respect to $\ln w_j$ ($j\neq i$). Only $\ln W$ depends on w_j:

$$\varepsilon_{ij}=-\frac{s_j}{1-v}+\sigma s_j=s_j\left(\sigma-\frac{1}{1-v}\right)=s_j\frac{\rho-v}{(1-\rho)(1-v)}.$$

Convert to a derivative: $\frac{\partial X_i^*}{\partial w_j}=\varepsilon_{ij}\frac{X_i^*}{w_j}$. Since $s_j/w_j=X_j^*/\sum_k w_k X_k^*$:

$$\frac{\partial X_i^*}{\partial w_j}=\frac{\rho-v}{(1-\rho)(1-v)}\cdot\frac{X_i^* X_j^*}{\sum_k w_k X_k^*}.$$

This expression is unchanged when i and j are swapped, so $\partial X_i^*/\partial w_j=\partial X_j^*/\partial w_i$. ✓ Inputs are gross substitutes if $\rho>v$ and gross complements if $\rho<v$.

Property 4: increasing in output price. Only the first bracket contains p:

$$\frac{\partial \ln X_i^*}{\partial \ln p}=\frac{1}{1-v}\Rightarrow\frac{\partial X_i^*}{\partial p}=\frac{X_i^*}{(1-v)p}>0.\checkmark$$

Property 5: negative semi-definite substitution matrix. The matrix of $\partial X_i^*/\partial w_j$ is minus the Hessian of the convex profit function shown in part (f), so it is negative semi-definite. ✓

Step 11: Properties of the output supply function. Take logs:

$$\ln Y^*=\frac{1}{1-v}\ln A+\frac{v}{1-v}\left(\ln v+\ln p-\ln W\right).$$

Property 1: homogeneous of degree zero. $v\,t\,p/(t\,W)=v\,p/W$, so $Y^*(t\,p,t\,w)=Y^*(p,w)$. ✓

Property 2: upward-sloping. $\frac{\partial \ln Y^*}{\partial \ln p}=\frac{v}{1-v}>0$ for $v<1$, so $\frac{\partial Y^*}{\partial p}=\frac{v}{1-v}\cdot\frac{Y^*}{p}>0$. ✓

Property 3: decreasing in input prices. $\frac{\partial \ln Y^*}{\partial \ln w_i}=-\frac{v}{1-v}s_i$, so $\frac{\partial Y^*}{\partial w_i}=-\frac{v}{1-v}\cdot\frac{s_i Y^*}{w_i}<0$. ✓

Property 4: reciprocity with input demand. Using $s_i/w_i=X_i^*/\sum_k w_k X_k^*=X_i^*/(v\,p\,Y^*)$:

$$-\frac{\partial Y^*}{\partial w_i}=\frac{v}{1-v}\cdot\frac{X_i^* Y^*}{v\,p\,Y^*}=\frac{X_i^*}{(1-v)p}=\frac{\partial X_i^*}{\partial p}.\checkmark$$

Answer (e): With $W=\left(\sum_i \alpha_i^{\sigma} w_i^{1-\sigma}\right)^{1/(1-\sigma)}$ and $\sigma=1/(1-\rho)$:

Output supply: $Y^*=A^{\frac{1}{1-v}}\left(\frac{v\,p}{W}\right)^{\frac{v}{1-v}}$. Input demand: $X_i^*=\left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}}\left(\frac{\alpha_i W}{w_i}\right)^{\frac{1}{1-\rho}}$.

All required properties are satisfied. Both functions are homogeneous of degree zero in (p,w). Input demands slope downward in own price and have symmetric cross-price effects. Supply slopes upward in p, and $\partial Y^*/\partial w_i=-\partial X_i^*/\partial p$.

(f) The Profit Function and Its Properties

Step 1: Substitute the optimal choices into profit.

$$\pi(p,w)=pY^*-\sum_i w_i X_i^*.$$

From part (e), Step 9, $\sum_i w_i X_i^*=v\,p\,Y^*$, so

$$\pi(p,w)=pY^*-v\,pY^*=(1-v)\,pY^*.$$

Step 2: Substitute Y^* and collect powers.

$$\pi(p,w)=(1-v)\,p\cdot A^{\frac{1}{1-v}}v^{\frac{v}{1-v}}p^{\frac{v}{1-v}}W^{-\frac{v}{1-v}}.$$

The powers of p combine as $p^1\cdot p^{\frac{v}{1-v}}=p^{\frac{1-v+v}{1-v}}=p^{\frac{1}{1-v}}$. Therefore

$$\pi(p,w)=(1-v)v^{\frac{v}{1-v}}(pA)^{\frac{1}{1-v}}W^{-\frac{v}{1-v}}.$$

Step 3: Write the profit function in compact form. Let $K=(1-v)v^{v/(1-v)}A^{1/(1-v)}>0$, $a=\frac{1}{1-v}>1$ and $b=\frac{v}{1-v}>0$. Then

$$\pi=K\,p^a W^{-b},\quad a-b=\frac{1-v}{1-v}=1.$$

Step 4: Check the properties.

Property 1: non-decreasing in p, non-increasing in w_i.

$$\frac{\partial\pi}{\partial p}=aK\,p^{a-1}W^{-b}>0,\quad \frac{\partial\pi}{\partial w_i}=-bK\,p^aW^{-b-1}\frac{\partial W}{\partial w_i}=-bK\,p^aW^{-b-1}\left(\frac{\alpha_i W}{w_i}\right)^{\sigma}<0.\checkmark$$

Property 2: homogeneous of degree one in (p, w). Since $W(t w) = t W(w)$,

$$\pi(t p, t w) = K(t p)^{a}(t W)^{-b} = K t^{a} p^{a} t^{-b} W^{-b} = t^{a-b}\pi(p, w) = t\,\pi(p, w). \checkmark$$

Property 3: convex in (p, w). Treat $g(p, W) = p^{a} W^{-b}$. Its second derivatives are

$$g_{pp} = a(a-1)p^{a-2}W^{-b} > 0,\; g_{WW} = b(b+1)p^{a}W^{-b-2} > 0,\; g_{pW} = -a\,b\,p^{a-1}W^{-b-1}.$$

The determinant of the Hessian is

$$det\, H = g_{pp}g_{WW} - g_{pW}^{2} = p^{2a-2}W^{-2b-2}\left[a(a-1)b(b+1) - a^{2}b^{2}\right].$$

Expanding the bracket:
$a(a-1)b(b+1) - a^{2}b^{2} = a\,b\left[(a-1)(b+1) - a\,b\right] = a\,b\left[a\,b + a - b - 1 - a\,b\right] = a\,b(a-b-1)$. Since $a - b = 1$, this equals 0:

$$det\, H = a\,b(a-b-1)\,p^{2a-2}W^{-2b-2} = 0.$$

The diagonal is positive and the determinant is zero, so H is positive semi-definite and g is convex in (p, W). Since g is decreasing in W and $W(w)$ is concave in w (it is a unit cost function), the composition $\pi(p, W(w))$ is convex in (p, w). ✓

Property 4: continuous in (p, w) for $p > 0$ and $w_i > 0$. ✓

Property 5: Hotelling's lemma. Differentiate with respect to p:

$$\frac{\partial \pi}{\partial p} = (1-v)v^{\frac{v}{1-v}}A^{\frac{1}{1-v}} \cdot \frac{1}{1-v}p^{\frac{1}{1-v}-1}W^{-\frac{v}{1-v}} = v^{\frac{v}{1-v}}A^{\frac{1}{1-v}}p^{\frac{v}{1-v}}W^{-\frac{v}{1-v}} = A^{\frac{1}{1-v}}\left(\frac{v\,p}{W}\right)^{\frac{v}{1-v}} = Y^{*}. \checkmark$$

Differentiate with respect to w_i, using $\partial W / \partial w_i = (\alpha_i W / w_i)^{\sigma}$:

$$-\frac{\partial \pi}{\partial w_i} = (1-v)v^{\frac{v}{1-v}}(p\,A)^{\frac{1}{1-v}} \cdot \frac{v}{1-v}W^{-\frac{v}{1-v}-1} \cdot \left(\frac{\alpha_i W}{w_i}\right)^{\sigma}.$$

Simplify: $(1-v) \cdot \frac{v}{1-v} = v$, then $v \cdot v^{\frac{v}{1-v}} = v^{\frac{1}{1-v}}$, and $W^{-\frac{v}{1-v}-1} = W^{-\frac{1}{1-v}}$. So

$$-\frac{\partial \pi}{\partial w_i} = v^{\frac{1}{1-v}}(p\,A)^{\frac{1}{1-v}}W^{-\frac{1}{1-v}}\left(\frac{\alpha_i W}{w_i}\right)^{\sigma} = \left(\frac{v\,p\,A}{W}\right)^{\frac{1}{1-v}}\left(\frac{\alpha_i W}{w_i}\right)^{\sigma} = X_i^{*}. \checkmark$$

Answer (f): $\pi(p, w) = (1-v)v^{\frac{v}{1-v}}(p\,A)^{\frac{1}{1-v}}\left[\sum_i \alpha_i^{\sigma} w_i^{1-\sigma}\right]^{-\frac{v}{(1-v)(1-\sigma)}}$, equivalently $\pi = (1-v)\,p\,Y^{*}$. It is increasing in p, decreasing in each w_i, homogeneous of degree one, convex and continuous, and it satisfies Hotelling's lemma. All the properties of a profit function hold.

(g) Two-Step Profit Maximisation

Profit maximisation can be split into two stages. First, find the cheapest way to produce any output level. Second, choose the output level that maximises profit given that cost.

Step 1: Cost minimisation (solved in part h). For any Y, the minimum cost and the conditional demands are

$$C(w,Y)=\left(\frac{Y}{A}\right)^{1/v}W=A^{-1/v}Y^{1/v}W,\ X_i^c(w,Y)=\left(\frac{Y}{A}\right)^{1/v}\left(\frac{\alpha_i W}{w_i}\right)^{\sigma}.$$

Step 2: Marginal cost. Differentiate C with respect to Y:

$$\mathrm{MC}(Y)=\frac{\partial C}{\partial Y}=\frac{1}{v}A^{-1/v}Y^{\frac{1}{v}-1}W=\frac{1}{v}A^{-1/v}Y^{\frac{1-v}{v}}W.$$

Step 3: Choose output to maximise $\pi(Y)=pY-C(w,Y)$.

$$\frac{d\pi}{dY}=p-\mathrm{MC}(Y)=0\Leftrightarrow p=\frac{1}{v}A^{-1/v}Y^{\frac{1-v}{v}}W.$$

The SOC is $\frac{d\,\mathrm{MC}}{dY}=\frac{1-v}{v^2}A^{-1/v}Y^{\frac{1-2v}{v}}W>0$, which holds when $v<1$, so marginal cost is rising.

Step 4: Solve for output. Multiply both sides by $vA^{1/v}/W$:

$$Y^{\frac{1-v}{v}}=\frac{vpA^{1/v}}{W}.$$

Raise both sides to the power $\frac{v}{1-v}$:

$$Y^*=\left(\frac{vpA^{1/v}}{W}\right)^{\frac{v}{1-v}}=\left(A^{1/v}\right)^{\frac{v}{1-v}}\left(\frac{vp}{W}\right)^{\frac{v}{1-v}}=A^{\frac{1}{1-v}}\left(\frac{vp}{W}\right)^{\frac{v}{1-v}}.$$

This matches the output supply from part (e).

Step 5: Recover the input demands. First compute $\left(Y^*/A\right)^{1/v}$:

$$\frac{Y^*}{A}=A^{\frac{1}{1-v}-1}\left(\frac{vp}{W}\right)^{\frac{v}{1-v}}=A^{\frac{v}{1-v}}\left(\frac{vp}{W}\right)^{\frac{v}{1-v}}=\left(\frac{vpA}{W}\right)^{\frac{v}{1-v}}\Rightarrow\left(\frac{Y^*}{A}\right)^{1/v}=\left(\frac{vpA}{W}\right)^{\frac{1}{1-v}}.$$

Substitute into the conditional demands:

$$X_i^*=X_i^c(w,Y^*)=\left(\frac{vpA}{W}\right)^{\frac{1}{1-v}}\left(\frac{\alpha_i W}{w_i}\right)^{\sigma}.$$

This is identical to the result from direct maximisation in part (e).

Step 6: Profit from the two-step route. $\pi = p\,Y^{*} - C\left(w, Y^{*}\right)$. At the optimum, $p = \mathrm{MC} = C/\left(v\,Y^{*}\right)$, so $C\left(w, Y^{*}\right) = v\,p\,Y^{*}$ and $\pi = (1-v)\,p\,Y^{*}$, the same as part (f).

> **Answer (g):** (1) Minimise the cost of producing Y to get $C\left(w, Y\right) = \left(Y/A\right)^{1/v} W$. (2) Choose Y so that $p = \mathrm{MC}\left(Y\right)$, which gives $Y^{*} = A^{1/(1-v)}\left(v\,p/W\right)^{v/(1-v)}$. Substituting Y^{*} into $X_i^{c}\left(w, Y\right)$ gives the same unconditional demands and profit as in parts (e) and (f), so the two procedures are equivalent. An interior optimum requires $v < 1$, so that marginal cost is increasing.

(h) Cost Minimisation and Conditional Input Demands

Step 1: The problem and the Lagrangian.

$$\min_{X_1, X_2, X_3} w_1 X_1 + w_2 X_2 + w_3 X_3 \text{ s.t. } A\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)^{v/\rho} = Y,$$

$$L = w_1 X_1 + w_2 X_2 + w_3 X_3 - \lambda\left[A\left(\alpha_1 X_1^{\rho} + \alpha_2 X_2^{\rho} + \alpha_3 X_3^{\rho}\right)^{v/\rho} - Y\right].$$

Step 2: First-order conditions.

$$\frac{\partial L}{\partial X_1} = w_1 - \lambda\, v\, A\, \alpha_1 X_1^{\rho-1} S^{\frac{v-\rho}{\rho}} = 0,$$

$$\frac{\partial L}{\partial X_2} = w_2 - \lambda\, v\, A\, \alpha_2 X_2^{\rho-1} S^{\frac{v-\rho}{\rho}} = 0,$$

$$\frac{\partial L}{\partial X_3} = w_3 - \lambda\, v\, A\, \alpha_3 X_3^{\rho-1} S^{\frac{v-\rho}{\rho}} = 0,$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow A\,S^{v/\rho} = Y \Rightarrow S = \left(\frac{Y}{A}\right)^{\rho/v}.$$

Step 3: Tangency conditions. Dividing the FOC for X_2 by the FOC for X_1 (the terms $\lambda\, v\, A\, S^{(v-\rho)/\rho}$ cancel):

$$\frac{w_2}{w_1} = \frac{\alpha_2}{\alpha_1}\left(\frac{X_2}{X_1}\right)^{\rho-1} \Rightarrow \left(\frac{X_2}{X_1}\right)^{\rho-1} = \frac{\alpha_1 w_2}{\alpha_2 w_1}.$$

Raise both sides to the power $\frac{1}{\rho-1} = -\sigma$:

$$\frac{X_2}{X_1} = \left(\frac{\alpha_1 w_2}{\alpha_2 w_1}\right)^{-\sigma} = \left(\frac{\alpha_2 w_1}{\alpha_1 w_2}\right)^{\sigma} \Rightarrow X_2 = X_1\left(\frac{\alpha_2 w_1}{\alpha_1 w_2}\right)^{\sigma}.$$

In the same way, from the FOCs for X_3 and X_1:

$$X_3 = X_1\left(\frac{\alpha_3 w_1}{\alpha_1 w_3}\right)^{\sigma}.$$

The general form is $X_i = X_1\left(\frac{\alpha_i w_1}{\alpha_1 w_i}\right)^{\sigma}$ (which also holds trivially for $i=1$).

Step 4: Substitute into the constraint. Compute each term $\alpha_i X_i^{\rho}$, using $\sigma\rho = \sigma - 1$:

$$\alpha_i X_i^{\rho} = \alpha_i X_1^{\rho}\left(\frac{\alpha_i w_1}{\alpha_1 w_i}\right)^{\sigma-1} = X_1^{\rho}\left(\frac{w_1}{\alpha_1}\right)^{\sigma-1}\alpha_i^{\sigma} w_i^{1-\sigma}.$$

Summing over $i = 1, 2, 3$:

$$S = X_1^{\rho}\left(\frac{w_1}{\alpha_1}\right)^{\sigma-1}\left(\alpha_1^{\sigma} w_1^{1-\sigma} + \alpha_2^{\sigma} w_2^{1-\sigma} + \alpha_3^{\sigma} w_3^{1-\sigma}\right) = X_1^{\rho}\left(\frac{w_1}{\alpha_1}\right)^{\sigma-1} W^{1-\sigma}.$$

Step 5: Solve for X_1. Set S equal to $(Y/A)^{\rho/\nu}$:

$$X_1^{\rho}\left(\frac{w_1}{\alpha_1}\right)^{\sigma-1} W^{1-\sigma} = \left(\frac{Y}{A}\right)^{\rho/\nu} \Rightarrow X_1^{\rho} = \left(\frac{Y}{A}\right)^{\rho/\nu}\left(\frac{\alpha_1}{w_1}\right)^{\sigma-1} W^{\sigma-1} = \left(\frac{Y}{A}\right)^{\rho/\nu}\left(\frac{\alpha_1 W}{w_1}\right)^{\sigma-1}.$$

Raise to the power $1/\rho$, using $(\sigma-1)/\rho = \sigma$:

$$X_1^c = \left(\frac{Y}{A}\right)^{1/\nu}\left(\frac{\alpha_1 W}{w_1}\right)^{\sigma}.$$

Step 6: The other inputs. Substitute X_1^c into $X_2 = X_1(\alpha_2 w_1/\alpha_1 w_2)^{\sigma}$:

$$X_2^c = \left(\frac{Y}{A}\right)^{1/\nu}\left(\frac{\alpha_1 W}{w_1}\right)^{\sigma}\left(\frac{\alpha_2 w_1}{\alpha_1 w_2}\right)^{\sigma} = \left(\frac{Y}{A}\right)^{1/\nu}\left(\frac{\alpha_2 W}{w_2}\right)^{\sigma},$$

since α_1 and w_1 cancel. Similarly, $X_3^c = \left(\frac{Y}{A}\right)^{1/\nu}\left(\frac{\alpha_3 W}{w_3}\right)^{\sigma}$.

Step 7: Cost function.

$$C(w, Y) = \sum_i w_i X_i^c = \left(\frac{Y}{A}\right)^{1/\nu} W^{\sigma}\sum_i w_i \alpha_i^{\sigma} w_i^{-\sigma} = \left(\frac{Y}{A}\right)^{1/\nu} W^{\sigma}\sum_i \alpha_i^{\sigma} w_i^{1-\sigma} = \left(\frac{Y}{A}\right)^{1/\nu} W^{\sigma} W^{1-\sigma} = \left(\frac{Y}{A}\right)^{1/\nu} W.$$

Step 8: Checks.

- **Shephard's lemma:** $\partial C/\partial w_i = (Y/A)^{1/\nu}\partial W/\partial w_i = (Y/A)^{1/\nu}(\alpha_i W/w_i)^{\sigma} = X_i^c$. ✓

- **Homogeneity:** $X_i^c(t\,w, Y) = (Y/A)^{1/v}\left(\alpha_i\, t\, W/(t\, w_i)\right)^{\sigma} = X_i^c(w, Y)$, so demands are homogeneous of degree zero in w. $C(t\,w, Y) = (Y/A)^{1/v}\, t\, W = t\, C(w, Y)$, so cost is homogeneous of degree one. ✓
- **Own-price effect:** $\ln X_i^c = \frac{1}{v}\ln\frac{Y}{A} + \sigma\left(\ln\alpha_i + \ln W - \ln w_i\right)$, so $\partial \ln X_i^c / \partial \ln w_i = \sigma\, s_i - \sigma = -\sigma\left(1 - s_i\right) < 0$. ✓
- **Constraint check:** $\sum_i \alpha_i \left(X_i^c\right)^{\rho} = (Y/A)^{\rho/v} W^{\sigma-1} \sum_i \alpha_i^{\sigma} w_i^{1-\sigma} = (Y/A)^{\rho/v} W^{\sigma-1} W^{1-\sigma} = (Y/A)^{\rho/v}$, so Y is produced exactly. ✓
- **Lagrange multiplier:** $\lambda = \partial C/\partial Y = \mathrm{MC} = \frac{1}{v} A^{-1/v} Y^{\frac{1-v}{v}} W$.

Answer (h): The conditional input demands are $X_i^c(w, Y) = \left(\frac{Y}{A}\right)^{1/v}\left(\frac{\alpha_i W}{w_i}\right)^{\frac{1}{1-\rho}}$, $i = 1, 2, 3$, where $W = \left(\sum_j \alpha_j^{\frac{1}{1-\rho}} w_j^{\frac{-\rho}{1-\rho}}\right)^{-\frac{1-\rho}{\rho}}$. The cost function is $C(w, Y) = (Y/A)^{1/v} W$.

Problem 5: Theft Insurance for Bob and Doug

Given. Initial wealth is \$2,500. A theft costs \$1,600, so wealth after a theft is \$900. Utility is $u(w) = \sqrt{w}$, which is strictly concave, so both travellers are risk-averse.

- Doug (high risk): theft probability $\pi_D = 1/2$.
- Bob (low risk, wears a money belt): theft probability $\pi_B = 1/4$.

The relevant utility values are $\sqrt{2500} = 50$ and $\sqrt{900} = 30$.

(a) Expected Utility without Insurance

Step 1: Expected utility formula.

$$\mathrm{EU} = \pi\, u(900) + (1-\pi)\, u(2500) = 30\,\pi + 50\,(1-\pi).$$

Step 2: Doug.

$$\mathrm{EU}_D = \frac{1}{2}(30) + \frac{1}{2}(50) = 15 + 25 = 40.$$

Step 3: Bob.

$$\mathrm{EU}_B = \frac{1}{4}(30) + \frac{3}{4}(50) = 7.5 + 37.5 = 45.$$

Step 4: Expected wealth (for later comparison).

$$E[w_D] = \frac{1}{2}(900) + \frac{1}{2}(2500) = \$1,700, \quad E[w_B] = \frac{1}{4}(900) + \frac{3}{4}(2500) = \$2,100.$$

Answer (a): Doug's expected utility is $\mathrm{EU}_D = 40$ and Bob's is $\mathrm{EU}_B = 45$. Bob is better off because the money belt lowers his probability of theft.

(b) Maximum Willingness to Pay for Full Insurance

With full insurance at premium P, wealth is certain at $2500 - P$. The maximum premium makes the traveller indifferent between insuring and not insuring:

$$\sqrt{2500 - P_{\max}} = \mathrm{EU} \Rightarrow P_{\max} = 2500 - (\mathrm{EU})^2.$$

Squaring both sides gives the certain wealth that yields the same utility (the certainty equivalent, CE): $2500 - P_{\max} = \mathrm{CE} = (\mathrm{EU})^2$.

Step 1: Doug.

$$\sqrt{2500 - P_D} = 40 \Rightarrow 2500 - P_D = 40^2 = 1600 \Rightarrow P_D^{\max} = \$900.$$

Step 2: Bob.

$$\sqrt{2500 - P_B} = 45 \Rightarrow 2500 - P_B = 45^2 = 2025 \Rightarrow P_B^{\max} = \$475.$$

Step 3: Decompose into expected loss and risk premium.

	Expected loss ($\pi \times 1600$)	Certainty equivalent	Risk premium	Maximum premium
Doug	\$800	\$1,600	\$100	\$900
Bob	\$400	\$2,025	\$75	\$475

Here the risk premium equals expected wealth minus the certainty equivalent: $1700 - 1600 = 100$ for Doug and $2100 - 2025 = 75$ for Bob. The maximum premium equals the expected loss plus the risk premium.

Answer (b): Doug will pay at most **\$900** and Bob at most **\$475** for full coverage of the \$1,600 loss. Each figure exceeds the traveller's expected loss (\$800 and \$400) because both are risk-averse.

(c) Actuarially Fair Insurance with Observable Types

An actuarially fair premium equals the expected payout, $P = \pi \times 1600$, so the insurer earns zero expected profit.

Step 1: Fair premiums.

$$P_D^{\text{fair}} = \frac{1}{2}(1600) = \$800, P_B^{\text{fair}} = \frac{1}{4}(1600) = \$400.$$

Step 2: Compare with willingness to pay.

$$\text{Doug: } 800 < 900 \Rightarrow \text{buys}; \text{Bob: } 400 < 475 \Rightarrow \text{buys}.$$

Step 3: Utility with insurance.

$$U_D = \sqrt{2500 - 800} = \sqrt{1700} \approx 41.23 > 40, U_B = \sqrt{2500 - 400} = \sqrt{2100} \approx 45.83 > 45.$$

Step 4: Gain to each buyer (consumer surplus in money terms).

$$\text{Doug: } 900 - 800 = \$100, \text{Bob: } 475 - 400 = \$75.$$

These are exactly the risk premia found in part (b): with fair insurance each traveller keeps the full value of removing risk.

Step 5: Insurer's side. Expected profit per policy equals premium minus expected payout:

$$E[\Pi_D) = 800 - \frac{1}{2}(1600) = 800 - 800 = 0, E[\Pi_B) = 400 - \frac{1}{4}(1600) = 400 - 400 = 0.$$

Each policy breaks even in expectation, so insurers are willing to offer both contracts.

Answer (c): Yes, a viable insurance market exists. Because insurers can tell the types apart, they charge Doug $800 and Bob $400. Each premium is below the buyer's maximum willingness to pay ($900 and $475), and each breaks even for the insurer. Both risk-averse travellers buy full insurance and are strictly better off. This is the efficient, full-insurance outcome.

(d) Pooled Premium when Types Are Unobservable (50% Bob, 50% Doug)

Step 1: Average probability of theft.

$$\pi_{\text{pool}} = \frac{1}{2}\left(\frac{1}{4}\right) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8} = 0.375.$$

Step 2: Actuarially fair pooled premium.

$$P_{\text{pool}} = \frac{3}{8} \times 1600 = \$600.$$

Step 3: Who buys at $600?

$$\text{Doug: } 600 < 900 \Rightarrow \text{buys}; \text{Bob: } 600 > 475 \Rightarrow \text{does not buy}.$$

Step 4: Insurer's expected profit if everyone bought at $600. For a pool of 100 people (50 of each type):

$$\text{Premiums} = 100 \times 600 = \$60,000,$$

$$\text{Expected claims} = 50 \times \frac{1}{2} \times 1600 + 50 \times \frac{1}{4} \times 1600 = 40,000 + 20,000 = \$60,000.$$

This would break even, but only if both types buy. Bob-types do not.

Step 5: Adverse selection. Once Bob-types leave, only Doug-types remain in the pool. At \$600 the insurer's expected payout per policy is $\frac{1}{2}(1600) = 800 > 600$, so it loses \$200 per policy. The \$600 pooled price therefore cannot be an equilibrium. The premium must rise until it breaks even on the remaining high-risk pool:

$$P^* = \pi_D \times 1600 = \$800.$$

With only the 50 Doug-types insured at \$600, premiums would be $50 \times 600 = \$30,000$ against expected claims of $50 \times \frac{1}{2} \times 1600 = \$40,000$, a loss of \$10,000. At \$800 premiums are $50 \times 800 = \$40,000$, matching expected claims.

At \$800 Doug still buys ($800 < 900$) and Bob still does not ($800 > 475$), so the outcome is self-consistent.

> **Answer (d):** The break-even pooled premium would be \$600, but Bob-types will not pay more than \$475, so they drop out. The market then contains only Doug-types, and the **equilibrium price is \$800**. Only high-risk (Doug) types are insured. Low-risk (Bob) types cannot buy insurance at a price they accept, even though they would buy at their fair price of \$400. This is adverse selection: the good risks are driven out of the market.

(e) Minimum Share of Low-Risk Types for Both to Buy

Let θ be the share of Bob-types (low risk) in the population.

Step 1: Average probability and pooled premium as functions of θ.

$$\pi_{\text{pool}}(\theta) = \theta \cdot \frac{1}{4} + (1-\theta) \cdot \frac{1}{2} = \frac{1}{2} - \frac{1}{4}\theta,$$

$$P_{\text{pool}}(\theta) = 1600\,\pi_{\text{pool}}(\theta) = 800 - 400\,\theta.$$

Check at $\theta = \frac{1}{2}$: $P_{\text{pool}} = 800 - 200 = 600$, matching part (d).

Step 2: Participation conditions.

- Doug buys if $P_{\text{pool}}(\theta) \le 900$. Since $P_{\text{pool}}(\theta) \le 800$ for every $\theta \in [0,1)$, this always holds.
- Bob buys if $P_{\text{pool}}(\theta) \le 475$. This is the binding condition.

Step 3: Solve Bob's condition.

$$800 - 400\theta \le 475 \Rightarrow 400\theta \ge 325 \Rightarrow \theta \ge \frac{325}{400} = \frac{13}{16} = 0.8125.$$

Step 4: Verify at $\theta = 0.8125$.

$$\pi_{\text{pool}} = 0.5 - 0.25(0.8125) = 0.296875,\ P_{\text{pool}} = 0.296875 \times 1600 = \$475 = P_B^{\max}. \checkmark$$

At this share Bob is exactly indifferent and Doug strictly prefers to buy, since $475 < 900$.

Step 5: Pooled premium for selected shares of Bob-types.

Share of Bob-types θ	$\pi_{\text{pool}} = 0.5 - 0.25$	Pooled premium $800 - 400\theta$	Doug buys? (≤ 900)	Bob buys? (≤ 475)
0.50	0.375	\$600	Yes	No
0.70	0.325	\$520	Yes	No
0.80	0.300	\$480	Yes	No
0.8125	0.296875	\$475	Yes	Yes (indifferent)
0.90	0.275	\$440	Yes	Yes
1.00	0.250	\$400	n/a	Yes

Answer (e): Bob-types must make up at least $\theta_{\min} = 13/16 = 81.25\%$ of the population. At that share the pooled fair premium falls to \$475, which equals Bob's maximum willingness to pay, so both low-risk and high-risk types buy insurance. With 50% Bob-types, as in part (d), the share is too low and the low-risk types drop out.